program 13

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13 . Write a R program to create multiple dot plots for grouped data comparing the distribution of variable across different categories using ggplot'2 position_dodge function.

Step 1: Load Required library

```
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Step 2: Load the dataset

```
head(ToothGrowth)
```

```
   len supp dose
1  4.2   VC  0.5
2 11.5   VC  0.5
3  7.3   VC  0.5
4  5.8   VC  0.5
5  6.4   VC  0.5
6 10.0   VC  0.5
```

```
summary(ToothGrowth) #provides summary statistics for each variable in the ToothGrowth dataset
```

```
      len        supp         dose
 Min.   : 4.20   OJ:30   Min.   :0.500
 1st Qu.:13.07   VC:30   1st Qu.:0.500
 Median :19.25           Median :1.000
 Mean   :18.81           Mean   :1.167
 3rd Qu.:25.27           3rd Qu.:2.000
 Max.   :33.90           Max.   :2.000
```

```
class(ToothGrowth)
```

```
[1] "data.frame"
```

```
class(ToothGrowth$dose)
```

```
[1] "numeric"
```

```
table(ToothGrowth$dose)
```

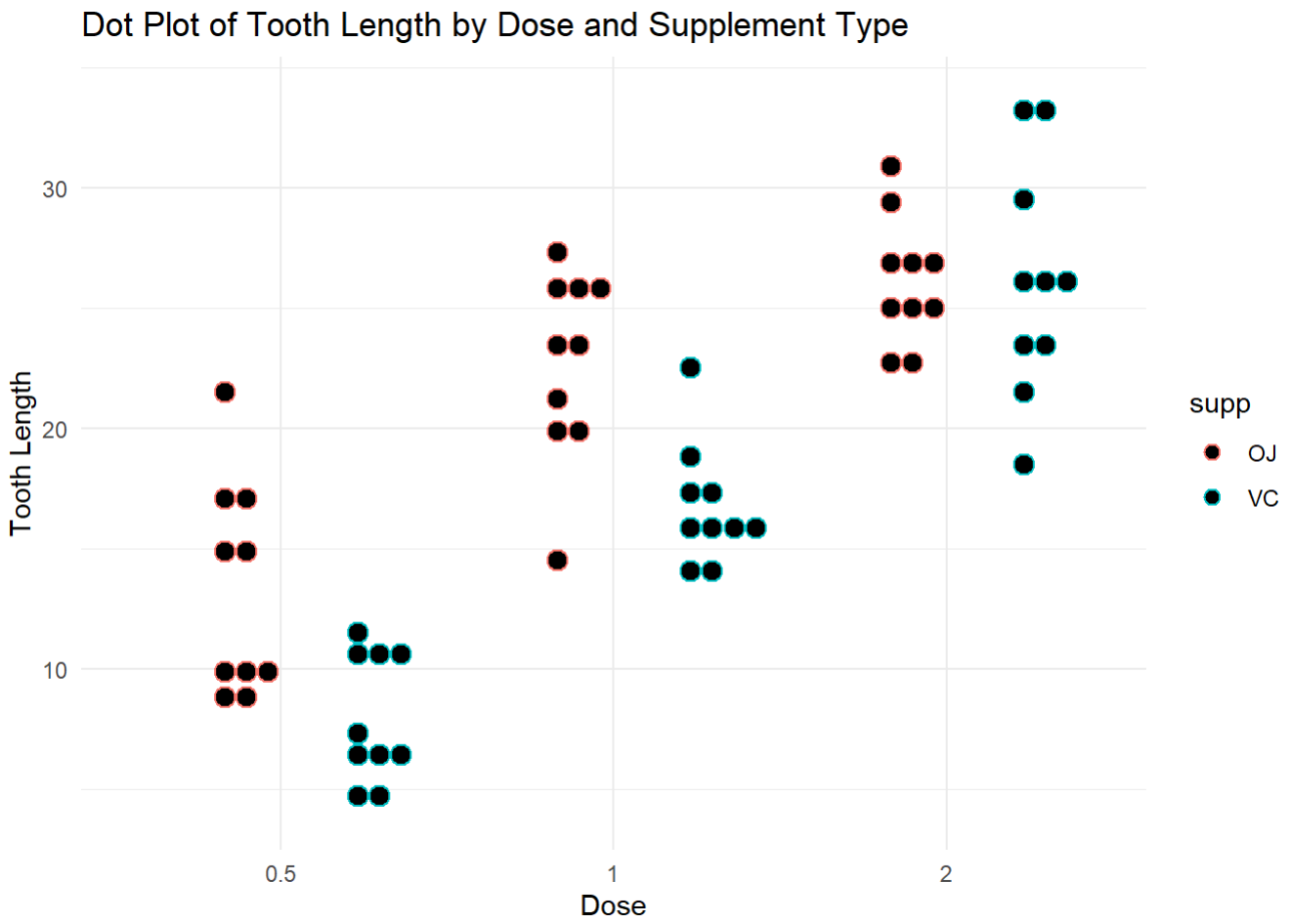
```
0.5   1   2
 20  20  20
```

Step 3: Create multiple dotplots

```
#Converts the dose column from a numeric variable to a factor, so it's treated as a categorica
ToothGrowth$dose <- as.factor(ToothGrowth$dose)

#Comment: Initializes the ggplot object using the ToothGrowth dataset.
ggplot(ToothGrowth,aes(x=dose,y=len,color=supp))+
    geom_dotplot(
      binwidth = 1.5, #controls the spacing of dots on y axis
    binaxis = "y",
    stackdir = "up",# centers the stracking of dots
    stackratio = 1,
    dotsize = 0.6,      # set the size of each dots
    position = position_dodge(width=0.8)

  ) +
  # Adds labels to the plot:
  labs(title = "Dot Plot of Tooth Length by Dose and Supplement Type",
       x = "Dose",
       y = "Tooth Length") +
  #for a clean and simple appearance
  theme_minimal()
```


Dot Plot of Tooth Length by Dose and Supplement Type
Tooth Length
30
20
10
0.5
1
2
Dose
supp
OJ
VC